

Chronic Disease Control, Insurance Coverage, and Healthcare Utilization Among California Adults: A Machine Learning Analysis of CHIS 2024

Authors and Information

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GitHub repository: <https://github.com/aparker03/chis-2024-health-outcomes-ml>

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1. Introduction

Chronic disease management and healthcare use reflect a mix of health needs, individual characteristics, and access to care. Andersen's Behavioral Model describes healthcare utilization in terms of predisposing characteristics, enabling resources, and need, including factors such as demographic characteristics, insurance, income, and having a regular source of care (Andersen, 1995). Prior CHIS studies have examined related patterns in diabetes management, cardiovascular disease and hypertension, and access to care across demographic and insurance groups (Choi et al., 2011; Tran et al., 2016; Babey et al., 2022). Machine-learning methods have also been applied to CHIS data to examine heterogeneity in survey responses (Ramirez et al., 2019).

Using the 2024 CHIS Adult Public Use File, this study evaluates how much demographic, socioeconomic, health, behavioral, and chronic-condition information can predict CHIS-defined chronic risk-factor control and annual doctor-visit utilization. It also tests whether insurance, affordability, and healthcare-access measures add predictive value and whether performance is consistent across race/ethnicity, poverty, insurance, and urban/rural groups. The models describe predictive patterns within the 2024 CHIS sample rather than causal effects or future clinical risk.

2. Research Questions

The project addresses four predictive research questions:

1. To what extent can demographic, socioeconomic, health, behavioral, and chronic-condition information classify CHIS-defined chronic risk-factor control when the direct measures used to determine control are excluded?
2. How much additional predictive value do insurance, healthcare-affordability, and access-to-care measures provide beyond the base characteristics for classifying CHIS-defined chronic risk-factor control?
3. To what extent can the available CHIS characteristics predict annual doctor-visit utilization, and how much additional predictive value is provided by insurance and healthcare-access measures?
4. How consistent is predictive performance across race/ethnicity, poverty, insurance, and urban/rural subgroups?

3. Dataset Description Including Attributes and Source

3.1 California Health Interview Survey

The analysis used the **2024 California Health Interview Survey (CHIS) Adult Public Use File**. The CHIS 2024 adult data consist of individual respondent records collected during the 2024 portion of the CHIS 2023–2024 survey cycle and are documented by the UCLA Center for Health Policy Research. The original adult public-use dataset contained **24,810 respondents and 897 variables**. The CHIS data dictionary provides variable names, descriptions, response categories, eligible respondent populations, and additional notes for interpreting the survey data. (CHIS, 2025).

For this project, variables were selected to represent demographic and socioeconomic characteristics, health behaviors, general health, chronic conditions, insurance coverage, healthcare affordability, and access to care. The final analysis focused on two prediction tasks: **(1) whether all applicable chronic risk factors were controlled** and **(2) the number of doctor visits reported during the past 12 months**.

3.2 Outcomes

Two outcomes were used in the machine-learning analysis:

Outcome	CHIS Variable	Definition
Chronic risk-factor control	RISK_CONTROL → not_controlled	CHIS defines RISK_CONTROL as whether all diagnosed chronic risk factors were under control during the past year. Respondents with valid values of 1 or 2 were retained and recoded: 0 = all applicable risk factors controlled; 1 = at least one applicable risk factor not controlled.
Healthcare utilization	AH5	Number of doctor visits reported during the past 12 months. This outcome was modeled as a continuous/count outcome for regression.

The project restricted the classification cohort to valid RISK_CONTROL values and created the binary not_controlled outcome. The final cohort included **8,560 controlled respondents and 3,140 respondents with at least one uncontrolled risk factor**. The CHIS data dictionary identifies RISK_CONTROL as “**All diagnosed chronic risk factors under control in past year.**”

3.3 Predictor Variables

The project compared two predefined predictor sets. The base set contained 17 variables representing demographic, socioeconomic, health, behavioral, and chronic-condition characteristics. The expanded set added seven insurance, affordability, and healthcare-access variables, for a total of 24 predictors. To reduce target leakage in the classification models, the direct CHIS measures used to determine the RISK_CONTROL outcome were excluded from the predictor sets. Thus, the models predicted chronic risk-factor control using related demographic,

socioeconomic, health, behavioral, chronic-condition, insurance, and healthcare-access characteristics rather than the underlying measures used to construct the target outcome.

Variable	Description
srage_p1	Respondent age
srsex	Self-reported gender/sex
racehp2_p1	Race/ethnicity
ur_clrt4	Urban/rural residence classification
sreduc	Educational attainment
wrkst_p1	Employment status
stablehouse	Whether the respondent has stable housing
ab1	Self-reported general health status
ac212	Physical activity measure
smkcur	Current smoking status
ab17	Asthma diagnosis/status
diabetes	Diabetes diagnosis/status
ab29v2	High blood pressure diagnosis/status
ab34	Heart disease diagnosis/status
ac6v2	History of stroke
povll2_p1v2	Poverty level relative to the federal poverty level
bmi_p	Body mass index (BMI)

The notebook explicitly identifies these variables as the project's base predictor set and labels them according to their analytic role. It also confirms that the base set contains 17 predictors. The CHIS data dictionary further identifies, for example, BMI_P as body mass index and POVLL2_P1V2 as poverty level expressed relative to 100% of the federal poverty level.

Expanded Insurance and Healthcare-Access Variables (7 Additional Variables)

Variable	Description
instype	Type of health insurance coverage
coverage_continuity	Continuity of health insurance coverage
usual	Whether the respondent has a usual place for healthcare
ah81b	Problems paying medical bills
ah22	Delayed or forgone needed healthcare
pc_ins	Difficulty finding a primary-care provider who accepts the respondent's insurance
aj218	Ease of obtaining needed care, tests, or treatment

These seven variables were added to test whether information about insurance, affordability, and access to care improved prediction beyond the 17-variable base set. The report identifies these exact expanded domains, and the notebook confirms the seven variables used to create the 24-variable expanded predictor set.

For example, the CHIS data dictionary describes PC_INS as difficulty finding a primary-care provider who accepts the respondent's insurance and AJ218 as how often it was easy to obtain needed care, tests, or treatment during the past six months.

3.4 Final Analytic Cohort

The analysis began with the **2024 CHIS Adult Public Use File containing 24,810 respondents and 897 variables**. Responses coded as refused, don't know, or not ascertained were treated as nonresponse, while structural skips were handled according to their survey meaning. After applying the outcome eligibility criteria and data-cleaning procedures, the final modeling cohort contained **11,700 adults**, with no missing values or duplicate respondent IDs.

The cohort was divided using a fixed split into 9,360 training respondents and 2,340 test respondents. The same train/test respondents were used across all four machine-learning model families to support consistent model comparisons, although the predictor sets varied by model: Decision Tree, Random Forest, and ANN workflows compared the 17-variable base and 24-variable expanded sets, while Logistic Regression, Linear Regression, and Elastic Net used only the expanded 24-variable set.

Race/ethnicity, poverty, insurance type, and urban/rural residence were also retained for subgroup evaluation. The CHIS survey weight was used for descriptive summaries rather than as a machine-learning predictor.

4. Model Building

Four model families were evaluated for each prediction task: regression-based models, decision trees, random forests, and artificial neural networks. Logistic Regression was used for classification, while Linear Regression and Elastic Net were used for regression. Decision Tree, Random Forest, and ANN models were developed for both outcomes. The tree-based and ANN workflows compared the 17-variable base set with the 24-variable expanded set, while the Logistic Regression, Linear Regression, and Elastic Net models used the expanded 24-variable set.

Model selection and tuning were performed with the training data before final evaluation on the held-out test set. Five-fold cross-validation was used to tune the Decision Tree, Random Forest, ANN, and Elastic Net models. Classification models accounted for the smaller not-controlled class through class weighting where applicable, and stratified folds were used for the classification tuning procedures. Regression models were evaluated on annual doctor visits, with the Random Forest and ANN workflows also comparing the original AH5 outcome with $\log_{1p}(\text{AH5})$ to address its right-skewed distribution. Final predictions from log-transformed models were converted back to the original visit scale before evaluation.

The final Decision Tree models used the expanded predictor set for both tasks. Random Forest training selected the expanded set for classification and regression, although the held-out classification results did not show an improvement over its base model. The ANN retained the base set for classification and the expanded set with $\log_{1p}(\text{AH5})$ for regression. Elastic Net was tuned across alpha and L1-ratio values and selected alpha = 0.01 and L1 ratio = 0.9, while Linear Regression served as the unregularized regression baseline.

5. Results and Evaluation of the Models

5.1 Classification Model Comparison

Model	Final Predictor Set	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression - Ivan	Expanded (24)	0.658	0.410	.600	0.490	0.712	0.464
Decision Tree - Krystal	Expanded	0.656	0.379	0.441	0.408	0.643	0.393
Random Forest - Sandipta	Expanded (24)	0.732	0.502	0.248	0.332	0.699	0.441
Artificial Neural Network - Alexis	Base (17)	0.655	0.409	0.642	0.500	0.703	0.464

The classification models showed different strengths rather than a single clear winner. The ANN was best at identifying respondents whose chronic risk factors were not controlled, while Logistic Regression provided the strongest overall separation between the controlled and not-controlled groups. Random Forest was the most accurate overall but missed many not-controlled respondents, showing that accuracy alone was not enough to judge performance for this outcome. The Decision Tree performed more modestly across the classification measures. Results from the predictor-set comparisons were also mixed, with added insurance and healthcare-access variables helping some models during selection but not consistently improving held-out classification performance.

5.2 Regression Model Comparison

Model	Final Predictor Set	Regression Approach	MAE	RMSE	R ²
Linear Regression - Ivan	Expanded (24)	Original AH5	4.131	7.861	0.078

Elastic Net - Ivan	Expanded (24)	Original AH5	4.109	7.846	0.082
Decision Tree - Krystal	Expanded	Original AH5	4.647	9.071	-0.227
Random Forest - Sandipta	Expanded (24)	log1p(AH5)	3.634	8.068	0.029
Artificial Neural Network - Alexis	Expanded (24)	log1p(AH5)	3.590	8.083	0.026

All regression metrics were calculated on the original AH5 scale. The Regression Approach column indicates whether each model was trained on AH5 directly or on log1p(AH5).

None of the regression models explained annual doctor-visit use particularly well. The ANN and Random Forest produced the smallest typical prediction errors, while Elastic Net and Linear Regression explained slightly more of the overall variation in visit counts. The Decision Tree performed worst and did not improve on a simple baseline in explaining variation. The log-transformed outcome helped the ANN and Random Forest handle the highly skewed visit distribution, but respondents with very high numbers of visits remained difficult to predict.

5.3 Subgroup Performance

Subgroup performance was not uniform across race/ethnicity, poverty, insurance, and urban/rural groups. The Logistic Regression, Elastic Net, Random Forest, and ANN results all showed some variation across these groups, although the size and direction of the differences depended on the model and outcome. A recurring pattern in the Random Forest and ANN classifiers was lower identification of not-controlled respondents in some higher-income and Asian subgroups, while regression error was generally higher for several Medicare and Medicaid groups than for respondents with employment-based insurance. Ivan's models also showed variation across the same subgroup dimensions, including differences in classification discrimination and doctor-visit prediction error.

Several of the smallest test subgroups produced less stable estimates. For example, the American Indian/Alaska Native group contained 43 respondents, the uninsured group contained 66, and Other Public insurance contained 25. Bootstrap intervals from the Random Forest and ANN analyses were wider for a number of these smaller groups. Subgroup results therefore show that aggregate performance does not fully describe how the models behaved across the test sample, but the differences should not be treated as evidence of causal disparities or model unfairness on their own.

6. Findings and Insights

6.1 CHIS-Defined Chronic Risk-Factor Control

The classification results show that broader CHIS characteristics contain some predictive information about the CHIS-defined control outcome even after the direct measures used to determine RISK_CONTROL were excluded. No model clearly dominated across every measure. The ANN identified more respondents in the not-controlled

group, while Logistic Regression provided the strongest overall discrimination. Adding insurance and healthcare-access variables did not consistently improve held-out classification performance, suggesting that these measures contributed less to this outcome than expected across the models that compared both predictor sets. Diagnosis variables such as asthma and diabetes appeared among the stronger predictors in some models, but they are structurally related to the CHIS-defined outcome and should be interpreted cautiously rather than as independent causes of control.

6.2 Healthcare Utilization

Annual doctor-visit use was harder to predict than CHIS-defined chronic risk-factor control. Across the regression models, prediction improved most for typical visit counts, while respondents with very high utilization remained difficult to estimate accurately. The expanded insurance and healthcare-access variables helped some regression models, especially the ANN and Random Forest, but overall explanatory power remained low. These results suggest that the available CHIS characteristics capture part of the utilization pattern, but much of the variation in annual doctor visits is tied to information not represented in the current predictor set.

6.3 Insurance and Healthcare Access

Insurance and healthcare-access measures added more useful information for predicting doctor-visit utilization than for classifying CHIS-defined chronic risk-factor control. The expanded predictor set was retained for regression in the Decision Tree, Random Forest, and ANN workflows, and access measures such as ease of obtaining needed care and having a usual place for healthcare appeared among the stronger regression predictors. Classification results were less consistent. The ANN performed better with the base set, Random Forest did not improve on the held-out test set after adding the access variables, and the Decision Tree showed only modest gains. These findings suggest that access-related information was more useful for predicting healthcare utilization than for classifying the CHIS-defined control outcome. Prior CHIS research has also shown that insurance type is associated with several dimensions of healthcare access, including delayed care, usual source of care, and provider or insurance acceptance (Babey et al., 2022).

6.4 Subgroup Patterns and Limitations

Predictive performance varied across race/ethnicity, poverty, insurance, and urban/rural groups, and smaller subgroups produced less stable estimates. These differences show that overall model performance does not fully describe how the models behaved across the test sample, but the subgroup results should not be interpreted as proof of bias, unfairness, or causal disparities. The study also relies on one year of largely self-reported CHIS data and has not been externally validated. CHIS survey weights were used for descriptive summaries but were not incorporated into model training, and prior work with complex survey data has shown that ignoring sampling weights can affect machine-learning performance and limit generalizability to the target population under some conditions (MacNeill et al., 2023).

7. References

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